# Md Obayed Reza Hridoy

+880 1924 043072 | 1obayedreza@gmail.com | linkedin.com/in/1obayed | github.com/1obayed

Electrical and Electronic Engineering student (BRAC University) and Computer Science associate student (University of the People), focused on embedded systems, embedded AI, and TinyML on resource-constrained hardware.

## Education

**BRAC University** — Dhaka, Bangladesh
*Bachelor of Science in Electrical and Electronic Engineering* — *2025 – Present*

**University of the People** — Online
*Associate Degree in Computer Science* — *2026 – Present*

## Experience

**Co-Founder, Head of Research & Management** — 2022 – Present
*Tornetec* — *Dhaka, Bangladesh*

- Leading research, strategic planning, and team management for an IT startup focused on AI integration into everyday human life.
- Overseeing prototype development and ensuring alignment with real-world technology needs.

**Undergraduate Researcher** — 2025 – 2026
*Laboratory of Space System Engineering Technology (LaSSET), BRAC University* — *Dhaka, Bangladesh*

- Working on computer vision and image-based detection tasks involving deep learning and CNN-based approaches.
- Developing and evaluating detection-based solutions for research applications using Python, MATLAB, Simulink, and PyTorch.

**Founder & Manager** — 2022 – 2024
*Zyphrst*

- Authorized NFT Selling Studio at Rarible Marketplace.
- Conducting in-depth market research to identify trends and opportunities in the NFT space.

## Projects

**GreenSync** | *Co-Founder, Systems Architect & Lead Researcher* — 2025 – Present

- Leading project planning, research coordination, and development of a technology-driven smart waste management system utilizing a reward-based user engagement model.
- Serving as the founding Project Manager and Co-Founder while contributing to ongoing research activities.
- University Innovation Hub Program (UIHP) & Hult Prize – Grant-Awarded Project.

**BRACU Alter** | *Co-Lead, AI & Vision Sub Team* — 2025 – 2026

- Working on deep learning and computer vision for image-based detection in disaster environments for rover navigation.
- Training and fine-tuning models for real-time multi-class object detection on embedded vision hardware.

## Achievements

**Champion, Ideation 5.0**

**Second Runner-Up, Hult Prize – BRAC University 2025**

**National Round Participant, Hult Prize Bangladesh**

## Technical Skills

**Languages**: C/C++, Python, Java
**Engineering Tools**: MATLAB, Simulink, PSpice
**AI & Computer Vision**: Computer Vision, PyTorch, Deep Learning, CNN, Image/Object Detection
**Research**: Data Analysis and Visualization, Quantitative Research Methods, Research Planning & Execution
**Professional**: Project Management, Interdisciplinary Collaboration, Strategic Planning